

Introduction to Demographic Methods

(1: Introduction)

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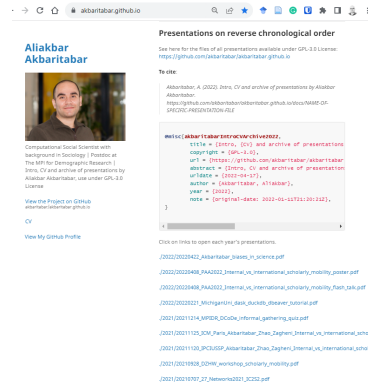
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Who am I?

- ▶ Computational Social Scientist with a background in sociology.
- ▶ Interest in science of science, scholarly migration, social and collaboration networks.
- ▶ Expertise in network analysis and modeling for large-scale and complex social media, bibliometric and textual data.



Practicing open science since 2011:
akbaritabar.github.io

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- Why did you join this course? What do you hope to learn?

Introduction to Demographic Methods



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- 1—• Introduction. Why study Demography? ... Git, version control, course materials GitHub
- 2—• Mortality
- 3—• Fertility
- 4—• Population projections
- 5—• Migration
- 6—• Kinship
- 7—• The future, and Digital and Computational Demography
- 8—• ?
- 9—• Wrap up + Description of examination
- 10—• Final examination: 15 min presentation to share your selected topic + Q&A; 6 weeks to prepare a short essay and empirical analysis

Learning and qualification objectives

- ▶ Estimation of basic demographic parameters.
- ▶ Capability to meaningfully interpret demographic parameters.
- ▶ Ability to critically evaluate demographic data and results that were published in the media.
- ▶ Ability to communicate demographic developments with scientists as well as with the general public.

Course content

In this course, we will study fundamental concepts, measures and models that demographers use to understand population dynamics. Demography as a discipline relies on a core methodological component that addresses problems related to:

- ▶ Measuring and estimating population rates;
- ▶ Developing models to expand our analytical thinking;
- ▶ Formalizing theories;
- ▶ Linking individual-level processes with macro-demographic trends;
- ▶ Producing forecasts that combine all the available information

Course description (with emphasized parts)

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Course/semester: Reading requirements

- Every week's materials include required and recommended readings specific to that week's topic, and some of the main ones are listed below.

1. Key material



Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.

2. Additional reading materials



Alexander, M. (2026).

Advanced Data Analysis course in Demographic Methods. Retrieved April 8, 2026, from <https://github.com/MJAlexander/soc6708>



Preston, S., Heuveline, P., & Guillot, M. (2000).

Demography: Measuring and Modeling Population Processes. Wiley.



Merli, M. G., Moody, J., Verdery, A., & Yacoub, M. (2023). Demography's Changing Intellectual Landscape: A Bibliometric Analysis of the Leading Anglophone Journals, 1950–2020. Demography, 10714127. <https://doi.org/10.1215/00703370-10714127>

- More information and the course's syllabus https://akbaritabar.github.io/Courses_Syllabus.html

Materials and sources that I will use during the semester

- ▶ The perks of teaching a course that is the most basic one in the field of Demography: 1) multiple text books (Preston et al., 2000; Wachter, 2014), 2) many publicly accessible materials
- ▶ Emilio Zagheni's 2018 course
- ▶ Monica Alexander's 2025-2026 course (Alexander, 2025, December 5/2026): <https://github.com/MJAlexander/soc6708>
- ▶ Alyson van Raalte's 2024 EDSO course
- ▶ U. California (UC) Berkeley's 2024 formal demography workshop (an additional focus on Migration)
- ▶ Ernesto Amaral's 2018 lecture
- ▶ LSHTM course on Introduction to Demographic Methods and Population Analysis <https://open.lshtm.ac.uk/mod/resource/view.php?id=7416>
- ▶ Applied Demography Toolbox: <https://applieddemogtoolbox.github.io>
- ▶ R packages that provide analysis tools/functions (Abel, 2025; Abel et al., 2024; Hyndman, Booth, et al., 2025; Hyndman, Tang, et al., 2025; Raalte, 2019, November 27/2025; Riffe, 2024, 2025; Riffe et al., 2025; Yeung et al., 2022)

Course's final examination

Active participation + preparing assignments during semester (25%).

6 weeks to prepare a final essay (guidelines and example will follow, 50%).

A 10 minutes presentation to deliver during the “last session of the course” + 5 minutes Q&A (25%).

The list of topics to select are those covered during the course. You will select one for your final essay and presentation.

What happens in “Hands-on part” and lab sessions?

Needed installations (or checking requirements)

R, RStudio, Renv virtual environments

Python (Vanilla or Miniforge), and virtualenv environments

Follow the Hands-On instructions posted in this weeks' folder in our GitHub repository ⇒

Required software for Hands-on sessions using R and/or Python

For the hands-on part of this week on web scraping and using Application Programming Interface (API)s, and following weeks, we will use R and Python.

It is necessary to have these two software installed and check if they are functioning e.g., by importing a package or checking their installed versions.

In Python, we will use [virtual environments](#) to isolate our projects and their required packages. This helps with reproducibility practices and making sure your code could be taken by someone else to re-run on their machine to recreate the results you have shown.

In R, if you are familiar with [Renv](#), feel free to use it to isolate your projects and R package requirements. The concepts are the same, but there are small differences in the details and syntax.

Install R and RStudio

The IT should already have installed R and RStudio on the lab computers.

If you are using R on Linux, I recommend using [R2U](#) to enable installing packages from binary instead of source that takes much less time.

To install it on your own, please follow these links:

- For Windows: <https://cran.r-project.org/bin/windows/base/>
- For Mac: <https://cran.r-project.org/bin/macosx/>
- For Linux: <https://cran.r-project.org/> Click on the button for Linux, then select your distribution, i.e., "Ubuntu" and follow the steps outlined to add it to Linux sources list and install using `sudo apt-get`

And once you installed R, visit this link to download and install RStudio: <https://posit.co/download/rstudio-desktop/>

Pyenv for Linux and Mac: Use Pyenv with vanilla, or conda, miniforge, etc Python versions on Linux or Mac "without" affecting the system installation of python

- This is a good introduction: <https://realpython.com/intro-to-pyenv/>
- Follow installation steps from this link (available for Mac and Linux, for Windows refer to a different fork, discussed in next section): <https://github.com/pyenv/pyenv>
- After the step on adding it to .bashrc, .bash_profile, and .profile, if you had an issue with bash not seeing it right i.e., if `pyenv --version` was giving an error like `/usr/bin/env: 'bash\r': No such file or directory`. You have to remove pyenv (go to the folder where it is installed i.e., `~/.pyenv` or something similar, and delete that folder), then edit `.git` configuration and set this from true to false i.e., `git`

Once you followed the guidelines and installed R, RStudio, and Python and checked with importing some libraries;

see each week's materials. It includes a "hands-on" folder with example scripts to use.

DISCLAIMER:

I will only share leads, links, and show you "example" scripts in R, and/or Python, in these introductory short sessions; you would need to extend them on your own to have a complete script/package to use for your final essay.

What to expect next?



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In addition to lectures, lab sessions with empirical work.

You need to learn Git and GitHub to access course materials.

How many of you know what Git is?

How many people have a GitHub account?

How many use R, Python, Stata, SQL, or something else?

For Git/GitHub, study the materials here

<https://github.com/akbaritabar/Using-Git-and-GitHub-for-Open-Science-Workshop>
and ask your questions.

How to access course's materials?

Create a GitHub profile, share the username with me to be added here:

https://github.com/akbaritabar/Course_introduction_to_demographic_methods_2026

Where to learn basics of R and Python?

To start with Python

Check this website first:

https://www.w3schools.com/python/python_intro.asp

Check this repository by Vincent Traag and others, for an introductory course and code:

<https://github.com/vtraag/intro-python>

Or this one by Data Carpentry:

<https://datacarpentry.org/python-ecology-lesson/>

To start with R

Check this website first:

<https://www.w3schools.com/r/default.asp>

This “very short introduction to R” is a good start:

<https://cran.r-project.org/doc/contrib/Torfs+Brauer-Short-R-Intro.pdf>

Or this course by Data Carpentry:

<https://datacarpentry.org/R-genomics/index.html>

Reading materials for the course (1/2)

Key reading material

-  Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.

Additional reading materials

-  Alexander, M. (2026). Advanced Data Analysis course in Demographic Methods. Retrieved April 8, 2026, from <https://github.com/MJAlexander/soc6708>
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Reading materials for the course (2/2)

R packages

-  Riffe, T. (2025). DemoTools. Retrieved December 16, 2025, from <https://timriffe.github.io/DemoTools/index.html>
-  Hyndman, R., Tang, S., McBain, M., & O'Hara-Wild, M. (2025). Vital: Tidy Analysis Tools for Mortality, Fertility, Migration and Population Data (Version 2.0.1). Retrieved December 16, 2025, from <https://cran.r-project.org/web/packages/vital/index.html>
-  Abel, G. J., K.C. S., Potancokova, M., Reiter, C., Tamburini, A., & Yildiz, D. (2024). Wcde: Download Data from the Wittgenstein Centre Human Capital Data Explorer (Version 0.0.7). Retrieved December 16, 2025, from <https://cran.r-project.org/web/packages/wcde/index.html>
-  Abel, G. J. (2025). Migest: Tools for Estimating, Measuring and Working with Migration Data (Version 2.0.5). Retrieved December 16, 2025, from <https://cran.r-project.org/web/packages/migest/index.html>
-  Hyndman, R., Booth, H., Tickle, L., Maindonald, J., Wood, S., & R Core Team. (2025). Demography: Forecasting Mortality, Fertility, Migration and Population Data (Version 2.0.1). Retrieved December 16, 2025, from <https://cran.r-project.org/web/packages/demography/index.html>
-  Raalte, A. van. (2025). Alysonvanraalte/Lifelneq. Retrieved February 7, 2026, from <https://github.com/alysonvanraalte/Lifelneq>
-  Riffe, T. (2024). DemoDecomp: Decompose Demographic Functions (Version 1.14.1). Retrieved December 16, 2025, from <https://cran.r-project.org/web/packages/DemoDecomp/index.html>
-  Riffe, T., Boe, C., Hilton, J., Goldstein, J., Holzman, S., & Yoo, S. H. (2025). HMDHFDplus: Read Human Mortality Database and Human Fertility Database Data from the Web (Version 2.0.8). Retrieved December 16, 2025, from <https://cran.r-project.org/web/packages/HMDHFDplus/index.html>
-  Yeung, J., Alexander, M., & Riffe, T. (2022). Rcbayes: Estimate Rogers-Castro Migration Age Schedules with Bayesian Models (Version 0.2.0). Retrieved December 11, 2024, from <https://cran.r-project.org/web/packages/rcbayes/>

Switch to Monica Alexander's slides: 1-intro

What to do before the next session?

1) Slides, Readings

1. Study my slides carefully.
2. Check the slide with reading materials.
3. Read the “key reading material”.
4. Prepare at least 1-2 discussion points, e.g., What was surprising and counterintuitive for you? What did you find that was already expected? What could be extended in this reading/work?
5. Skim over the additional materials.

2) Hands-on

1. Check the slide and folder for hands-on materials.
2. Install the requirements as outlined.
3. Run the codes in hands-on materials to reproduce the examples, and if you can extend them.
4. Note down the unclear points, errors, and questions for the next session.

Thanks for your attention!

Questions and comments are always welcome!

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subject line: “Demogr-SoSe-2026”

LinkedIn/BlueSky/Twitter/Mastodon: @akbaritabar

References

- Abel, G. J. (2025). Migest: Tools for Estimating, Measuring and Working with Migration Data (Version 2.0.5). Retrieved December 16, 2025, from <https://cran.r-project.org/web/packages/migest/index.html>
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- Preston, S., Heuveline, P., & Guillot, M. (2000). Demography: Measuring and Modeling Population Processes. Wiley.
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